

Table of Contents

Part 1: The Big Picture	2
The High Leverage Zone	2
Part 2: The Eight Groups	3
Group 1: Fundamentals	3
Group 2: Data & Knowledge	5
Group 3: Intelligence Layer	7
Group 4: Models & Providers	8
Group 5: Infrastructure & Connectivity	10
Group 6: Agents & Automation	12
Group 7: No-Code Builder Tools	13
Group 8: Business Layer	15
Part 3: Compounds	17
Compound 1: The Custom AI Assistant	17
Compound 2: The Automated Lead Pipeline	17
Compound 3: The Content Factory	18
Part 4: Quick Reference	18

Part 1: The Big Picture

The Periodic Table of AI Elements organises 48 essential AI concepts into 8 logical groups, mirroring how chemical elements build the physical world. Just as hydrogen and oxygen combine to form water, AI elements like Tokens, Prompts, and Models combine to form business systems. Understanding these elements individually is the first step; understanding how they combine is where real business value emerges.

The table is not just a taxonomy. It is a map of leverage. Groups 1 through 4 represent the foundations: what AI is and how it works. Groups 5 through 7 represent the connectors and builders: how you make AI work for you. Group 8 ties everything back to business outcomes. The highest entrepreneurial value concentrates in Groups 5, 6, and 7, what we call the High Leverage Zone, because this is where AI transforms from a tool you visit into a system that works for you around the clock.

OPTITUDE AI
The Periodic Table of AI Elements

Group 1	Group 2	Group 3	Group 4	Group 5	Group 6	Group 7	Group 8
1 Tk Token	7 Td Training Data	13 Sp System Prompt	19 Gp GPT	25 Ap API	31 Ag Agent	37 Zp Zero-Party	43 Uc Use Case
2 Md Model	8 Em Embedding	14 Mm Memory	20 Cl Chain	26 Wh Workflow	32 Or Orchestration	38 Mk Market	44 RI ROI
3 Pr Prompt	9 Vd Vector DB	15 Mu Multimodal	21 Gm Gemini	27 Ep EdgeOps	33 WF Workflow	39 N8 No-Code	45 Pe Prompt Eng.
4 Cw Context Window	10 Kb Knowledge Base	16 Rh RAG	22 LI LLM	28 Mc MCP	34 Ma Multi-Agent	40 Vf VoiceFlow	46 As AI Stack
5 Tp Temperature	11 Rg RAG	17 Gr Guardrails	23 Ms Model	29 Fc Function Calling	35 Hi Human-in-Loop	41 Fw Flowise	47 Av AI Avatars
6 Pm Parameters	12 Ft Fine-Tuning		24 Gk GenAI	30 Sk SDK	36 Tu Tool Use	42 Cr Custom	48 St AI Strategy

48 Elements | 8 Groups | 1 Complete Framework

Figure 1: The Complete Periodic Table of AI Elements

The High Leverage Zone

Groups 5 (Infrastructure and Connectivity), 6 (Agents and Automation), and 7 (No-Code Builder Tools) form the High Leverage Zone. These are the elements that let entrepreneurs build AI systems without needing to be AI researchers. If you want to create business value with AI, this is where you focus your time and energy. The foundations matter, but the leverage is here.

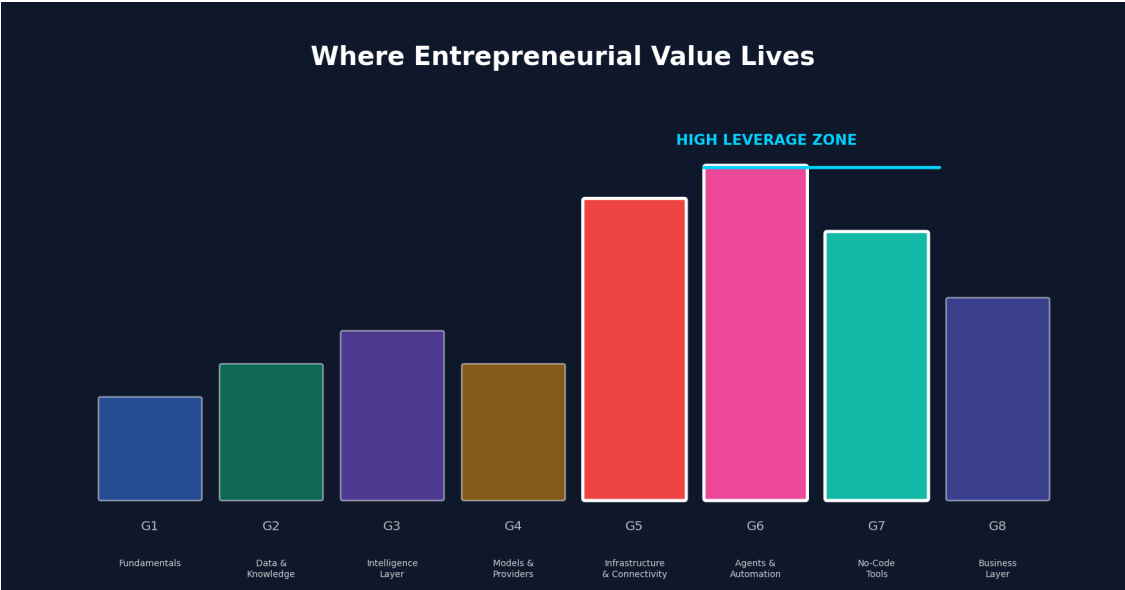


Figure 2: Where Entrepreneurial Value Lives

Part 2: The Eight Groups

Each group contains related elements that work together. Think of them as families in the periodic table: elements in the same group share characteristics and tend to combine in similar ways. We will explore each group in detail, with every element broken down into its symbol, definition, real-world example, key combinations, common mistakes, and a reflection question to test your understanding.

Group 1: Fundamentals

These are the atoms of everything. Nothing works without these. You will encounter every element in this group in literally every AI tool you ever use.



Figure 3: Group 1 Elements

Tk

Token

Token How AI reads text, not words. ~1 token per 0.75 words. Directly affects cost, speed, and how much you can feed an AI at once.

Real-World Example: A 1,000-word email is ~1,333 tokens. At \$0.03/1K tokens, that costs ~\$0.04.

Combines with: Context Window, Model, Prompt

Common mistakes: Confusing tokens with words; Ignoring token costs in automations; Not truncating inputs to fit context windows

Reflection: Can you estimate the token cost of sending a 5-page document to an AI?

Md

Model

Model The core AI engine. Different models have different strengths, costs, and capabilities. Choosing the right one is fundamental.

Real-World Example: Models are like vehicle engines: a city car for short trips, a truck for heavy loads. Don't use a truck to pop to the shops.

Combines with: Parameters, Temperature, Fine-Tuning

Common mistakes: Using the most expensive model for every task; Assuming all models handle every task equally; Not testing smaller models first

Reflection: Can you name three AI models and when you would choose each?

Pr

Prompt

Prompt Your instructions to the model. Structure determines ~80% of output quality. The most underrated skill for non-technical AI builders.

Real-World Example: 'Write a marketing email' gives anything. 'Write a 150-word email to SaaS founders promoting a 20% AI discount, friendly-professional tone, clear CTA by Friday' gives precision.

Combines with: System Prompt, Temperature, RAG

Common mistakes: Being vague and expecting precise results; Not iterating on prompts when output is poor; Blaming the model when the prompt is the issue

Reflection: Can you rewrite a vague prompt into a specific one and see the difference?

Cw

Context Window

Context Window How much the model can see in a single conversation. Like short-term memory with a hard limit.

Real-World Example: Imagine reading a book but only remembering the last 10 pages. GPT-4's 128K window holds ~96,000 words before things fade.

Combines with: Token, Memory, RAG

Common mistakes: Stuffing too much into one conversation; Not understanding why AI 'forgets'; Assuming bigger windows solve all problems

Reflection: If your AI keeps forgetting instructions mid-conversation, which element do you investigate?

Tp

Temperature

Temperature Controls creativity vs. predictability. Low = precise, consistent. High = creative, unpredictable.

Real-World Example: Legal contract clauses? Temperature near 0. Brainstorming marketing slogans? Crank to 0.8+ for wild ideas.

Combines with: Prompt, Model, Guardrails

Common mistakes: Using high temperature for automated processes; Expecting identical outputs above temperature 0; Not adjusting temperature when switching task types

Reflection: What temperature for an AI drafting customer emails vs. one writing poetry?

Pm Parameters	Parameters Internal settings that shape how a model thinks. More parameters = more capability AND more cost. Real-World Example: Llama 3 8B handles everyday tasks cheaply. GPT-4 handles complex reasoning at higher price. Combines with: Model, Token, Fine-Tuning Common mistakes: Assuming bigger is always better; Not comparing cost-effectiveness across sizes; Ignoring parameter counts when estimating API costs Reflection: Why might a 7B model actually be the better choice for some tasks?
-------------------------------------	--

Checkpoint: You should now be able to: Explain tokens, models, prompts, context windows, temperature, and parameters; Estimate token costs for a given task; Choose the right model and temperature for different use cases

Group 2: Data & Knowledge

How you feed information into an AI system and make it smarter for your specific use case. Without these, your AI only knows what it was trained on.

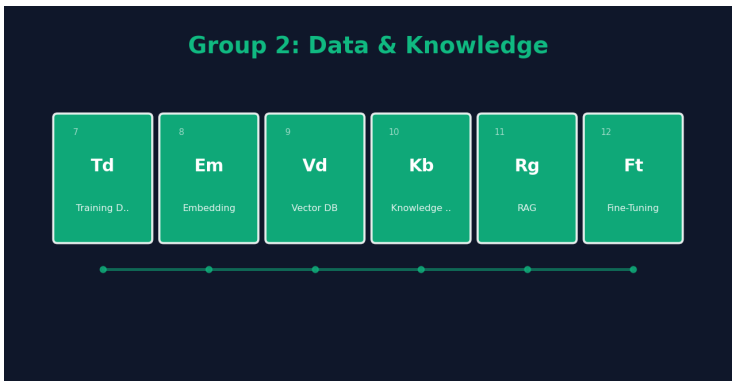


Figure 4: Group 2 Elements

Td Training Data	Training Data The information a model learned from during creation. Tells you where it is strong and where it has blind spots. Real-World Example: If trained on data up to Jan 2024, it literally cannot tell you about anything after that. Combines with: Fine-Tuning, RAG, Embedding Common mistakes: Expecting AI to know things outside its training data; Not checking cutoff dates; Assuming training data is always accurate Reflection: If AI gives wrong info about your company, is that a model problem or a training data problem?
--	--

Em

Embedding

Embedding How AI converts text into numbers that capture meaning. Enables semantic understanding.

Real-World Example: Like a library shelved by meaning, not alphabet. Search 'grow sales' and it finds 'revenue expansion' too.

Combines with: Vector Database, RAG, Training Data

Common mistakes: Confusing embeddings with keyword matching; Not understanding why semantic search beats text search; Ignoring embedding quality in search systems

Reflection: Why might an embedding search find a document with none of your search words?

Vd

Vector Database

Vector Database Where embeddings are stored and searched. If you build any AI that searches information, you need a vector database.

Real-World Example: A super-smart filing cabinet: search by meaning, not filename.

Combines with: Embedding, Knowledge Base, RAG

Common mistakes: Using a traditional database when vector DB is needed; Not indexing data before storing; Assuming vector DBs work like keyword search

Reflection: If your AI can never find the right document, which element might be misconfigured?

Kb

Knowledge Base

Knowledge Base A curated store of information your AI can search in real time. Unlike training data, it can be updated anytime.

Real-World Example: Update a price, and the AI immediately gives the correct answer.

Combines with: Vector Database, RAG, Guardrails

Common mistakes: Dumping unstructured data without organising; Not keeping the KB up to date; Expecting AI to know things not in the KB

Reflection: If AI gives outdated info, do you retrain the model or update the knowledge base?

Rg

RAG

RAG Retrieval Augmented Generation. Combines a knowledge base + model so AI answers using YOUR information.

Real-World Example: Without RAG: generic answers. With RAG: AI searches your handbook, retrieves the section, then generates an answer from your specific policy.

Combines with: Knowledge Base, Vector Database, Embedding

Common mistakes: Building chatbots without RAG and expecting business knowledge; Confusing RAG with fine-tuning; Not testing retrieval quality before deploying

Reflection: Can you explain the difference between an AI that knows general info and one that knows YOUR info?

Ft

Fine-Tuning

Fine-Tuning Training an existing model on your specific data to change how it behaves. More advanced than RAG, but powerful for specialised outputs.

Real-World Example: Need an AI that always writes in your brand's exact voice? Fine-tuning on hundreds of past communications achieves that.

Combines with: Training Data, Model, Parameters

Common mistakes: Using fine-tuning when RAG would suffice; Fine-tuning on too little data; Not evaluating fine-tuning vs. RAG first

Reflection: If you need AI to sound exactly like your brand every time, would you choose RAG or fine-tuning?

Checkpoint: You should now be able to: Distinguish training data from a knowledge base; Explain when to use RAG vs. fine-tuning; Understand embeddings and vector databases

Group 3: Intelligence Layer

The layer between the raw model and what it produces. Making AI smarter, more contextual, and safer to deploy in a real business.



Figure 5: Group 3 Elements

Sp System Prompt	System Prompt Hidden instruction set given before any conversation starts. Gives AI its persona, role, rules, and tone. Real-World Example: A customer service AI's system prompt: 'You are a helpful support agent. Be concise, friendly, never make up features.' Combines with: Prompt, Guardrails, Memory Common mistakes: Leaving system prompt empty = generic responses; Making it too long and contradictory; Not testing edge cases where AI might ignore it Reflection: If your AI keeps using the wrong tone, which element do you adjust first?
Mm Memory	Memory AI's ability to remember things across conversations. Without memory, every interaction starts from zero. Real-World Example: Without: you tell the AI your business name and goals every time. With: you tell it once, and every future visit it already knows you. Combines with: Context Window, Knowledge Base, RAG Common mistakes: Building chatbots with no memory = impersonal experience; Storing too much and hitting window limits; Not giving users control over what AI remembers Reflection: Would customers prefer an AI that remembers them or starts fresh each time?
Mu Multimodal	Multimodal AI that works with images, audio, and video, not just text. Business applications expanding fast. Real-World Example: Upload a kitchen photo: AI identifies style, suggests improvements, writes a listing description. Combines with: Model, Embedding, Agent Common mistakes: Assuming all models process images; Not compressing images before sending; Ignoring multimodal when it could solve your problem Reflection: What task in your business would be easier if AI could see images?

Rh RLHF	RLHF Reinforcement Learning from Human Feedback. The technique behind why modern AI sounds natural and helpful. Real-World Example: Without RLHF, 'How do I bake a cake?' might dump Wikipedia chemistry. With RLHF: a friendly step-by-step recipe. Combines with: Training Data, Guardrails, Model Common mistakes: Confusing RLHF with prompting techniques; Not understanding why AI sometimes refuses requests; Assuming RLHF makes AI perfectly safe Reflection: Why might an AI refuse to answer a question? Which element is responsible?
Gr Guardrails	Guardrails Safety boundaries and rules controlling what AI will and will not do. For customer-facing AI, this is not optional. Real-World Example: Banking AI without guardrails: 'Invest your life savings in crypto!' With guardrails: 'Never give specific investment advice.' Combines with: System Prompt, Human-in-the-Loop, RAG Common mistakes: Deploying customer AI without guardrails; Making guardrails too restrictive; Not testing with adversarial inputs Reflection: If AI said something inappropriate to a customer, do you blame the model or the guardrails?

Checkpoint: You should now be able to: Write effective system prompts; Design AI with appropriate guardrails; Understand RLHF and multimodal capabilities

Group 4: Models & Providers

The engine room: the actual AI models you build on. Most people spend all their time here but should not. Models are important, but just one piece of the system.

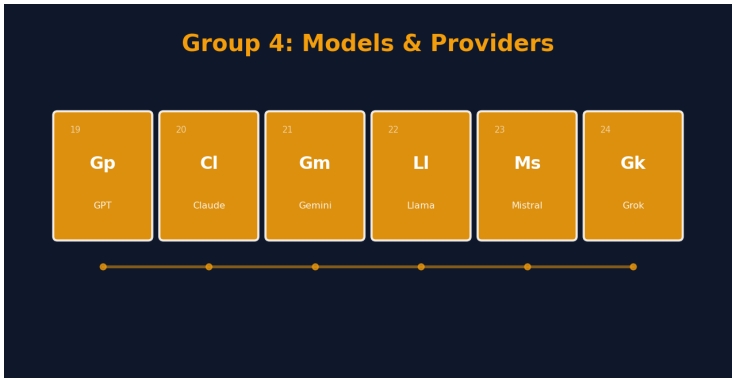


Figure 6: Group 4 Elements

Gp GPT	GPT OpenAI's versatile flagship. Excels at general-purpose tasks. GPT-4 for strong reasoning; GPT-4o-mini for cost-effective everyday tasks. Real-World Example: Customer support chatbot: GPT-4 gives strong reasoning. Simple FAQ: GPT-4o-mini at a fraction of the cost. Combines with: API, SDK, Context Window Common mistakes: Using GPT-4 for everything when mini suffices; Not understanding GPT-4 vs. GPT-4o vs. mini; Assuming GPT is always best Reflection: Can you name a task where GPT would NOT be the best choice?
Cl Claude	Claude Anthropic's model. Exceptional at long documents and nuanced reasoning. Context windows up to 200K tokens. Real-World Example: Analyse a 50-page legal contract for risks: feed the entire document at once and get thorough analysis. Combines with: MCP, Context Window, API Common mistakes: Using Claude where speed matters more than nuance; Not leveraging the long context window; Assuming Claude can not handle creative tasks Reflection: What task benefits most from reading an entire 50-page document at once?
Gm Gemini	Gemini Google's model with deep Google Workspace integration. If your business runs on Google, Gemini offers seamless connectivity. Real-World Example: Your team lives in Google Workspace. Gemini reads your Docs, analyses Sheets, drafts Gmail emails. Combines with: API, Multimodal, Model Common mistakes: Overlooking Gemini if you are in Google's ecosystem; Not exploring its multimodal capabilities; Assuming it is just another ChatGPT Reflection: If your business uses Google Workspace daily, how could Gemini's integration help?
Ll Llama	Llama Meta's open-source model. Run it yourself, modify it, deploy on your own infrastructure. Full control, no vendor lock-in. Real-World Example: Healthcare company can not send patient data to OpenAI. With Llama on their own servers, data never leaves. Combines with: Fine-Tuning, Parameters, Model Common mistakes: Assuming open-source means lower quality; Underestimating hosting infrastructure costs; Not considering Llama when privacy is required Reflection: When would running your own AI model on your own servers be worth the effort?
Ms Mistral	Mistral Lean, efficient European alternative. Strong performance at smaller sizes. Excellent value per dollar. Real-World Example: The efficient sedan of AI: not the flashiest, but excellent value per mile. Combines with: Parameters, API, Model Common mistakes: Dismissing Mistral because it is less famous; Not benchmarking against bigger models; Overlooking for European data compliance Reflection: When might a smaller, efficient model outperform a larger one for your use case?

Gk

Grok

Grok xAI's model with real-time X (Twitter) data access. Unique advantage for trend analysis and sentiment monitoring.

Real-World Example: Know what people say about your brand right now, not last month.

Combines with: API, Training Data, Model

Common mistakes: Assuming Grok is just another chatbot; Not leveraging real-time data access; Relying on it for deep accuracy over speed

Reflection: If you needed real-time brand sentiment, which model and why?

Checkpoint: You should now be able to: Compare GPT, Claude, Gemini, Llama, Mistral, and Grok; Choose the right model for a given business need; Understand open-source vs. proprietary trade-offs

Group 5: Infrastructure & Connectivity

The connectors that make AI actually useful in the real world. Without these, AI is just a website you visit. With these, it is a system that works for you.

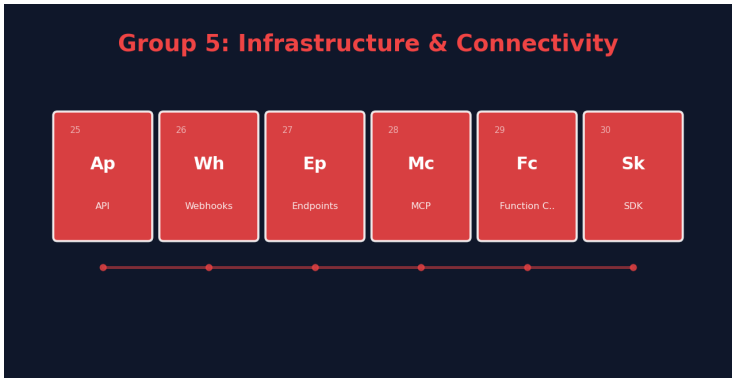


Figure 7: Group 5 Elements

Ap

API

API Application Programming Interface. The bridge between AI models and everything else. A way to call an AI model programmatically.

Real-World Example: Customer fills in a contact form: API call sends their message to AI, gets a response, sends it back in under 2 seconds.

Combines with: Endpoints, SDK, Function Calling

Common mistakes: Confusing APIs with the AI model itself; Not handling API errors and rate limits; Hard-coding API keys in source code

Reflection: Can you explain the difference between chatting with AI on a website vs. calling it through an API?

Wh

Webhooks

Webhooks One system automatically notifies another when something happens. The webhook is the starting gun.

Real-World Example: Customer completes a purchase on Shopify. Webhook fires to Make.com, triggers AI workflow: personalised thank-you email, CRM update, follow-up task.

Combines with: Workflows, API, Endpoints

Common mistakes: Using polling instead of webhooks for real-time events; Not securing webhook endpoints; Assuming webhooks guarantee delivery

Reflection: If your AI automation never starts, do you check the prompt or the webhook?

Ep

Endpoints

Endpoints The specific URL address where an API lives. Like a phone number: dial the right one and you are connected.

Real-World Example: When connecting tools like Make or Zapier, you constantly encounter endpoints.

Combines with: API, Webhooks, SDK

Common mistakes: Confusing endpoints with the full API; Not updating endpoints when API versions change; Hard-coding endpoint URLs

Reflection: If your integration stops after an API update, which element is likely the cause?

Mc

MCP

MCP Model Context Protocol. An emerging standard allowing AI models to connect to external tools consistently. Think: universal power plug for AI.

Real-World Example: Before MCP, every AI-tool connection was custom-built. MCP creates one universal standard: plug your agent into any MCP tool and it just works.

Combines with: Function Calling, API, Agent

Common mistakes: Ignoring MCP because it is new; Confusing MCP with a specific tool rather than a protocol; Not exploring MCP-compatible tools

Reflection: Why could a universal AI-tool standard transform how agents are built?

Fc

Function Calling

Function Calling The ability for AI to call external tools or APIs during a conversation. This is what lets AI actually DO things, not just say things.

Real-World Example: 'Book me a meeting with Sarah next Tuesday.' Function calling enables: check calendar, call booking API, send confirmation email.

Combines with: API, MCP, Agent

Common mistakes: Expecting AI to take actions without function calling; Not defining functions clearly enough; Forgetting to handle function call errors

Reflection: If your AI can only generate text but never does anything, what is missing?

Sk

SDK

SDK Software Development Kit. Pre-built libraries making it easier to build with AI APIs.

Real-World Example: Instead of 50 lines of raw HTTP code to call an AI model, the OpenAI SDK does it in 5 lines.

Combines with: API, Endpoints, Function Calling

Common mistakes: Writing raw API calls when an SDK exists; Not keeping SDKs updated; Assuming you can not use SDKs if you are not a developer

Reflection: Why is using an SDK like using a recipe kit instead of sourcing every ingredient?

Checkpoint: You should now be able to: Explain APIs, webhooks, and endpoints; Understand function calling and MCP; Know when to use SDKs vs. direct API calls

Group 6: Agents & Automation

Where AI stops being a tool you use and becomes a system that works for you. This is the biggest leap in the entire table. From chatbot to employee.



Figure 8: Group 6 Elements

Ag Agent	Agent An AI system that perceives its environment, makes decisions, and takes action. The leap from tool to digital employee. Real-World Example: Customer emails about an order. Agent looks up the order, sees it is delayed, checks courier API, drafts personalised response, offers discount. All without human intervention. Combines with: Orchestration, Tool Use, Workflows Common mistakes: Calling any chatbot an 'agent'; Building agents without guardrails or oversight; Expecting agents to be perfect from day one Reflection: What is the difference between a chatbot that answers questions and an agent that solves problems?
Or Orchestration	Orchestration The system that manages and coordinates multiple agents or steps. If an agent is a musician, orchestration is the conductor. Real-World Example: Content pipeline: research agent gathers data, writing agent drafts, editing agent polishes, publishing agent posts. Orchestration ensures the right order and failure handling. Combines with: Multi-Agent, Workflows, Human-in-the-Loop Common mistakes: Building multiple agents without orchestration; Not handling failure scenarios; Over-engineering when simple workflows suffice Reflection: If three agents needed to work together, what goes wrong without orchestration?
Wf Workflows	Workflows The sequence of automated steps producing a business outcome. Build workflows, not one-off prompts. Real-World Example: Lead comes in: 1) AI scores it, 2) drafts personalised email, 3) queued for review, 4) if approved, sent automatically, 5) CRM updated. 30 minutes becomes 30 seconds. Combines with: Webhooks, Agent, Human-in-the-Loop Common mistakes: Automating processes you have not manually tested; Building too-rigid workflows; Not monitoring performance after deployment Reflection: Can you identify one repetitive process that could become a workflow?

Ma Multi-Agent System	Multi-Agent System Multiple specialised AI agents working together. One researches, one writes, one edits. This replaces entire processes, not just single tasks. Real-World Example: Real estate: Agent 1 searches listings, Agent 2 analyses market data, Agent 3 generates comparison reports, Agent 4 drafts client emails. Combines with: Agent, Orchestration, Tool Use Common mistakes: Building one do-everything agent instead of specialists; Not defining clear boundaries between agents; Needing more robust error handling Reflection: If building an AI team for your business, what specialised agents would you create?
Hi Human-in-the-Loop	Human-in-the-Loop The point where a human reviews or approves before automation continues. Not a weakness but a design choice. Real-World Example: AI drafts refund responses. Before any email goes out, human reviews and clicks Approve. Over time, auto-approve small refunds, only flag large ones. Combines with: Workflows, Guardrails, Agent Common mistakes: Automating everything without checkpoints; Making review too burdensome; Not gradually reducing oversight as system proves reliable Reflection: At what point in your workflows would you insist on human approval?
Tu Tool Use	Tool Use An agent's ability to use external tools: search engines, calculators, databases, apps. Without tool use, an agent is just a fast chatbot. Real-World Example: Agent with tool use can: search web for stock prices, calculate compound interest, look up CRM records, send Slack messages. Combines with: Function Calling, MCP, API Common mistakes: Building agents without connecting tools; Giving agents tools without proper permissions; Not defining which tools for which tasks Reflection: If an agent could only generate text, not search/calculate/send, how useful would it really be?

Checkpoint: You should now be able to: Distinguish chatbots from agents; Design workflows with human-in-the-loop; Plan multi-agent systems with orchestration

Group 7: No-Code Builder Tools

The builder tools that let you create AI systems without writing code. If you are not a developer, this is your workshop. Every element above becomes drag-and-drop.

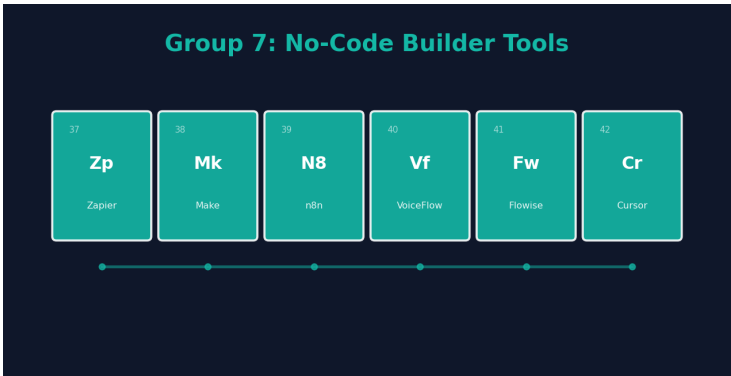


Figure 9: Group 7 Elements

Zp Zapier	Zapier The original no-code automation platform. 6,000+ app connections, low learning curve. Real-World Example: Set up a Zap: new lead fills form, add to CRM, send welcome email, notify team in Slack. No code. Combines with: Webhooks, API, Workflows Common mistakes: Using Zapier for complex multi-branch logic; Not monitoring task usage (costs scale); Assuming Zapier can do everything a custom build can Reflection: What is one repetitive daily task Zapier could automate in 10 minutes?
Mk Make	Make Built for complex, multi-step workflows. Visual, flexible, handles the logic that real AI systems require. Real-World Example: Build a scenario: receive webhook on order, check purchase history, AI generates upsell recommendation, send email, log in analytics. Combines with: Webhooks, API, Workflows Common mistakes: Using Zapier when Make's complexity is needed; Not using Make's error handling features; Building overly complex scenarios hard to maintain Reflection: If your automation needs branching logic, which tool do you reach for?
N8 n8n	n8n Open-source automation. More technical than Zapier/Make, but runs on your own infrastructure. Full control, no per-task pricing. Real-World Example: Process customer data through AI without sending it to third parties. n8n on your server: data never leaves. Combines with: API, Webhooks, Workflows Common mistakes: Choosing n8n without technical skills for self-hosting; Not budgeting for server costs; Underestimating maintenance burden Reflection: When would self-hosted automation be worth the extra technical effort?
Vf VoiceFlow	VoiceFlow No-code platform for building AI agents and conversational interfaces. Drag-and-drop design. Real-World Example: Customer service chatbot: drag conversation blocks, connect to knowledge base, set up fallbacks, design the journey visually. Deploy to your website in minutes. Combines with: Agent, Knowledge Base, Guardrails Common mistakes: Using VoiceFlow for non-conversational automations; Not testing with real users; Overcomplicating conversation design Reflection: Could you design a simple customer service flow on paper? That is what VoiceFlow lets you build digitally.
Fw Flowise	Flowise Open-source, no-code builder for LLM applications. Build RAG pipelines, custom agents, chatbots with a visual interface. Real-World Example: Chatbot that knows your documents: visually connect document loader, text splitter, embedding model, vector database, LLM. Each is a drag-and-drop block. Combines with: RAG, Vector Database, Agent Common mistakes: Confusing Flowise with a general automation tool; Not understanding underlying concepts before building visually; Self-hosting without proper security Reflection: To build a chatbot that knows your documents, what elements would you chain together?

Cr Cursor	Cursor AI-powered code editor. Describe what you want in plain English and it writes the code. Real-World Example: Need a custom API endpoint? In Cursor: 'Create a Python API that takes customer feedback, sends it to GPT-4 for categorisation, and returns the category.' Combines with: SDK, API, Model Common mistakes: Assuming Cursor outputs are always correct; Not learning basic concepts to evaluate output; Using Cursor when no-code would solve it faster Reflection: If you could describe a feature in English and have it built, what would you create?
---------------------	--

Checkpoint: You should now be able to: Choose between Zapier, Make, and n8n; Build conversational AI with VoiceFlow; Create RAG pipelines with Flowise; Use Cursor for custom builds

Group 8: Business Layer

The business layer where everything connects back to value. If it does not serve a use case, deliver ROI, or fit your strategy, it is a toy, not a tool.



Figure 10: Group 8 Elements

Uc Use Case	Use Case The specific business problem you are solving with AI. Must come before picking tools. Start with the problem, not the product. Real-World Example: Bad: 'We need to use AI.' Good: 'Support spends 20 hrs/week on 15 repeated questions. AI handles 80% = 16 hrs saved.' Combines with: ROI, AI Strategy, AI Stack Common mistakes: Starting with a tool instead of a problem; Defining use cases too vague to measure; Building solutions for problems that do not exist Reflection: Can you write three specific AI-solvable problems with measurable success criteria?
-----------------------	---

Ri ROI	ROI Return on Investment. The only real reason to implement AI. If you can not articulate time saved, revenue added, or cost reduced, rethink before you build. Real-World Example: Automate lead qualification: before 2 hrs/day manual sorting, after 15 min reviewing AI-screened leads. 1.75 hrs/day x 220 work days = 385 hrs/year. At \$50/hr = \$19,250 annual savings. Combines with: Use Case, AI Strategy, Workflows Common mistakes: Building AI without defining ROI measurement; Counting only time saved, not quality improved; Not tracking actual ROI after implementation Reflection: How would you calculate the exact dollar value of automating one process?
Pe Prompt Engineering	Prompt Engineering The craft of designing prompts that reliably produce the output you need. Highest-leverage thing a non-technical AI builder can learn. Real-World Example: 'Write a sales email' vs. 'Write 200-word email to SaaS founders whose teams waste 10+ hrs/week on manual data entry. Highlight 3-day free trial. Confident not pushy. CTA to book demo.' Combines with: Prompt, System Prompt, Temperature Common mistakes: Treating it as one-time instead of iterative; Not documenting successful prompts; Believing it replaces the need for good data and systems Reflection: Can you take a prompt you use regularly and improve it with specific constraints?
As AI Stack	AI Stack Your total combination of AI tools, models, and systems working together. Like a tech stack but for intelligence. Real-World Example: Claude for documents, GPT-4o-mini for chat, Pinecone for vectors, Make for workflows, VoiceFlow for chatbot, Google Sheets for data. Each piece has a job. Combines with: Model, No-Code Tools, Infrastructure Common mistakes: Adopting tools randomly without understanding fit; Having overlapping tools doing the same thing; Not documenting so team does not know what is where Reflection: Can you map every AI tool you use? Are there gaps or redundancies?
Av AI Avatars	AI Avatars Digital personas built on AI. Growing fast for content creation, customer service, brand building. Real-World Example: Fitness brand creates AI avatar of their head coach. Customers chat 24/7 for advice. One coach, infinite availability. Combines with: Model, System Prompt, Memory Common mistakes: Thinking avatars are just chatbots with a profile picture; Not investing in personality and consistency; Ignoring ethical considerations Reflection: Could your business benefit from an AI rep that is always available to customers?
St AI Strategy	AI Strategy Deliberate, sequenced plan for how AI integrates into your business over time. Without strategy: random tool collection. With strategy: genuine competitive advantage. Real-World Example: Random: try ChatGPT one week, build a chatbot next. Strategic: Phase 1 automate lead qualification, Phase 2 deploy support agent, Phase 3 multi-agent content system. Each phase builds on the last. Combines with: Use Case, ROI, AI Stack Common mistakes: Confusing 'using AI tools' with having an AI strategy; Not setting timelines and milestones; Building capabilities that do not connect to business goals Reflection: If you wrote a 12-month AI strategy, what would be your first three priorities?

Checkpoint: You should now be able to: Define specific, measurable AI use cases; Calculate ROI for AI implementations; Design a phased AI strategy; Map your AI stack

Part 3: Compounds

Individual elements are powerful, but their true value emerges when they combine into compounds. In chemistry, hydrogen and oxygen form water. In AI, Tokens, Prompts, Models, RAG, Vector Databases, and Workflows combine to form a Custom AI Assistant. Understanding compounds is how you move from knowing about AI to building with AI.

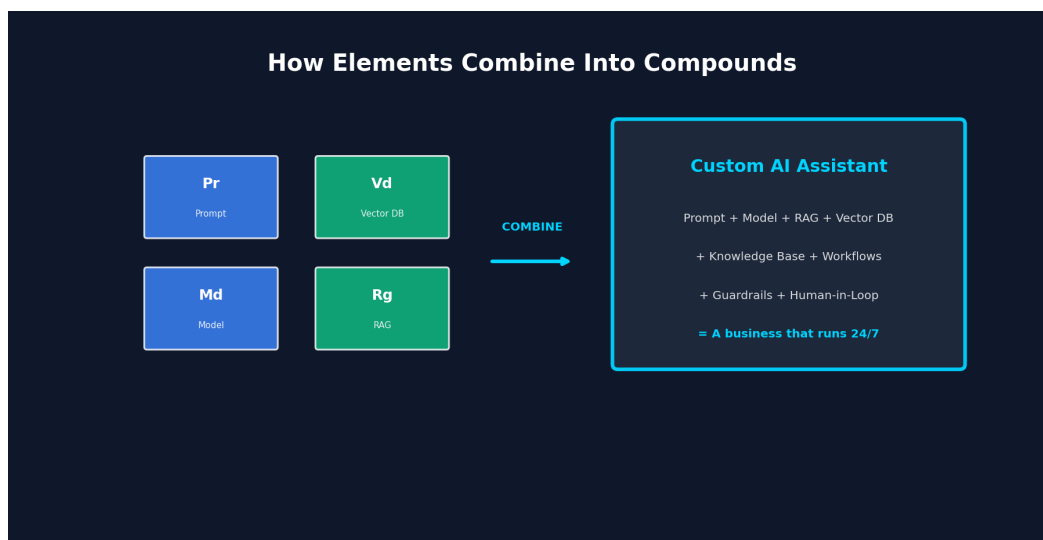


Figure 11: How Elements Combine Into Compounds

Compound 1: The Custom AI Assistant

Formula: Token + Prompt + Model + RAG + Vector DB + Knowledge Base + Guardrails + System Prompt

This is the most common compound in AI business. A custom AI assistant that knows your company's information, speaks in your brand's voice, and stays within safe boundaries. It is the backbone of customer support, internal helpdesks, and sales assistants. Without RAG, the AI only knows general information. Without guardrails, it might say something damaging. Without a system prompt, it has no consistent personality. Each element plays an essential role in making the compound work reliably.

Compound 2: The Automated Lead Pipeline

Formula: Webhook + Workflow + Agent + Function Calling + API + Human-in-the-Loop

A lead fills in a form. A webhook triggers a workflow. An AI agent scores the lead using function calling to check the CRM via API. If the score is high, it drafts a personalised email. A human reviews (Human-in-the-Loop), approves, and the email sends automatically. What used to take 30 minutes of manual work now takes 30 seconds,

with higher quality and zero missed follow-ups. This compound turns AI from a conversation tool into a revenue engine.

Compound 3: The Content Factory

Formula: Multi-Agent + Orchestration + Prompt Engineering + Make/n8n + Knowledge Base + AI Strategy

A research agent gathers data from your knowledge base. A writing agent drafts content using prompt engineering best practices. An editing agent polishes for brand voice. Orchestration coordinates the sequence. Make or n8n handles the automation triggers. Your AI strategy ensures the content aligns with business goals. The result: scaled, consistent, high-quality content production that would require an entire team to match manually.

Part 4: Quick Reference

Use this section as a rapid lookup for any element. The table below lists all 48 elements with their symbol, group, and a one-line definition. Refer back to the detailed element cards in Part 2 for full descriptions, examples, and combinations.

Sym	Element	Group	Definition
Tk	Token	G1	How AI reads text; affects cost/speed
Md	Model	G1	The core AI engine
Pr	Prompt	G1	Your instructions; 80% of output quality
Cw	Context Window	G1	How much the model can remember at once
Tp	Temperature	G1	Creativity vs. predictability control
Pm	Parameters	G1	Internal settings shaping model thinking
Td	Training Data	G2	What the model learned during creation
Em	Embedding	G2	Converts text to meaning-capturing numbers
Vd	Vector Database	G2	Stores and searches embeddings
Kb	Knowledge Base	G2	Curated info store AI can search in real time
Rg	RAG	G2	Combines knowledge base + model for your data
Ft	Fine-Tuning	G2	Trains a model on your specific data
Sp	System Prompt	G3	Hidden instructions setting AI personality
Mm	Memory	G3	Remembers across separate conversations
Mu	Multimodal	G3	AI that works with images, audio, video
Rh	RLHF	G3	Why AI sounds natural (human feedback training)
Gr	Guardrails	G3	Safety boundaries for AI behaviour
Gp	GPT	G4	OpenAI's versatile model family
Cl	Claude	G4	Anthropic's model; great for long docs
Gm	Gemini	G4	Google's model; ecosystem integration
Ll	Llama	G4	Meta's open-source model; full control

Sym	Element	Group	Definition
Ms	Mistral	G4	Lean, efficient European alternative
Gk	Grok	G4	xAI's model; real-time social data
Ap	API	G5	Bridge between AI and other systems
Wh	Webhooks	G5	Automatic notifications between systems
Ep	Endpoints	G5	URL address where an API lives
Mc	MCP	G5	Universal standard for AI-tool connections
Fc	Function Calling	G5	Lets AI call external tools and take action
Sk	SDK	G5	Pre-built libraries for easier AI building
Ag	Agent	G6	AI that perceives, decides, and acts
Or	Orchestration	G6	Conductor coordinating multiple agents
Wf	Workflows	G6	Sequence of automated steps for outcomes
Ma	Multi-Agent System	G6	Multiple specialised agents working together
Hi	Human-in-the-Loop	G6	Human review checkpoint in automation
Tu	Tool Use	G6	Agent's ability to use external tools
Zp	Zapier	G7	No-code automation; 6,000+ apps
Mk	Make	G7	Visual builder for complex workflows
N8	n8n	G7	Open-source automation; self-hosted
Vf	VoiceFlow	G7	No-code builder for conversational AI
Fw	Flowise	G7	Open-source visual builder for LLM apps
Cr	Cursor	G7	AI-powered code editor; describe and build
Uc	Use Case	G8	Specific business problem you're solving
Ri	ROI	G8	Return on investment; the real reason
Pe	Prompt Engineering	G8	Craft of designing effective prompts
As	AI Stack	G8	Total combination of AI tools and systems
Av	AI Avatars	G8	Digital persona built on AI
St	AI Strategy	G8	Deliberate plan for AI business integration

A practical AI consultancy for UK businesses.

Optitude AI helps organisations apply AI intelligently, commercially, and safely. We cut through the hype to focus on the implementations that actually move the needle — time saved, manual workload reduced, decisions made with better information, and AI adoption that is structured rather than ad-hoc.

We work across the full adoption lifecycle, from the first “where do we start?” conversation through to embedded, governed, day-to-day AI use inside live workflows. Our starting point is always the business problem, never the tool. We are equally comfortable advising a founder who has never logged into ChatGPT as we are working with an operations director who already has a stack of AI licences and needs to make them pay back.

What we do

AI readiness reviews

A structured assessment of where your business is today and what would need to be true before AI creates reliable value.

AI opportunity mapping

Identifying and prioritising the highest-leverage use cases across your organisation.

Workflow and process analysis

Mapping current processes so AI is fitted to how you actually work, not the other way around.

AI productivity improvement

Targeting measurable time savings in admin-heavy, repeatable tasks.

AI training for teams and leaders

Practical, role-relevant sessions that build confidence and competence without requiring technical backgrounds.

AI policy and governance support

Safe-usage policies, data handling principles, and approval workflows that reduce risk from unstructured AI use.

Recruitment and HR process improvement

Applying AI to sourcing, screening, onboarding and people analytics with appropriate human oversight.

Practical AI implementation support

Hands-on help selecting tools, configuring workflows, and embedding AI into real day-to-day operations.

A reminder of what you now hold.

You now have a complete map of the AI landscape — 48 elements across eight groups, from the foundations of tokens and models, through data and intelligence layers, infrastructure, agents, no-code builders, and the business layer where ROI actually lives. That map is not the end of the journey. It is the starting point for a conversation about where AI could create real value inside your specific business.

Reading the guide is the first step. The next is to identify two or three high-leverage use cases — places where AI could plausibly save time, reduce manual workload, or enable capabilities your team cannot reach today. The eight groups in the table give you the vocabulary to scope those use cases precisely. The compounds section shows you how individual elements combine into the systems a real business actually runs.

What working with Optitude AI typically delivers

Save time — on repeatable, admin-heavy work that does not require human judgement.

Improve decision-making — with faster access to summarised, relevant information.

Better people decisions — AI-assisted sourcing, screening, and analytics with human oversight.

Clearer AI adoption plans — sequenced, measured, aligned to business goals — not scattered.

Reduce manual workload — across operations, HR, finance, and client communication.

Increase productivity — in roles where output is currently capped by manual throughput.

Stronger business development — faster, more consistent lead qualification and content production.

Reduced AI risk — addressing shadow AI: staff pasting sensitive data into public tools.

If you have read this far, you already have enough context to start the conversation. The offer on the next page is for you.

EXCLUSIVE OFFER FOR GUIDE DOWNLOADERS

A free AI Opportunity Review.

Thirty minutes. No obligation. Yours because you downloaded this guide.

The Periodic Table of AI Elements gives you the framework. The AI Opportunity Review turns it into a plan. In a focused 30-minute session, an Optitude AI consultant will walk through your business with you, identify two or three high-leverage use cases, and outline what implementation would actually look like — timeline, effort, expected return, and risk.

- Two or three high-leverage AI use cases specific to your business
- A clear picture of timeline, effort, and expected return for each
- An honest view on what to do first — and what to wait on

BOOK YOUR FREE REVIEW

periodic-table.optitudeai.com/#book-a-review

Or reply “review” to any email from Optitude AI and we will book you in directly.

Optitude AI · Practical AI consultancy for UK businesses · optitude360.com